Loïc Strullu 2023

Practical Work 3: Crop rotations with perennial crop

I. First part

We want to simulate a crop rotation on the long-term corresponding to a treatment of the ORE ACBB of the INRAe in the Picardie region in northern France. The winter wheat simulated in previous practical works is a part of this rotation. The experiment starts the 19 February 2009.

Description of the crop rotation:

The crop rotation is presented in the file inputs_STICS_PW3.xlsx in the tab USMs. We are going to look at it together.

Spring barley – spring pea – winter wheat – rapeseed – spring barley – corn – winter wheat – alfalfa for 2 years – winter wheat – spring barley – corn – winter wheat

All the spring crop are preceded by a catch crop: mustard, crimson clover or vetch.

Description of the model initialization:

The model initialization is presented in the tab Ini.

One initialization file for the crop rotation (ACBB_Mons_T5_ini.xml) beginning in bare soil.

Two initialization files for alfalfa simulation: necessity to activate the option code_acti_reserve because this option is activated in the plant file and necessity to have an initialization at stage “lev” for chaining alfalfa simulations.

Description of the technical practices:

Information about technical practices is summarized in tab Tec. We are going to look at it together.

No exogenous organic residues.

Soil tillage corresponds to weed control, preparation of soil for sowing or deep tillage for catch crop destruction (be careful, in STICS, a tillage won’t destroy a crop!!!!). Residues management (ressuite) with restitution: straw + roots or stubble + roots for cash crops; roots for alfalfa, whole_crop for catch crop.

Description of sowing (date, depth, density, variety) and harvest date (here you can destroy your catch crop).

Dates, amount and type of mineral fertilizer used. N fertilization is rather low!

Finally, you have a description of cutting management for alfalfa. With Code_hautfauche_dyn = 2 the model will simulate alone the residual biomass and leaf area index in function of cutting height (here hautcoupe is parameterized at 7 centimeters). I recommend to you to use this option!

The station and the soil are the same that we used for previous practical works 1 and 2.

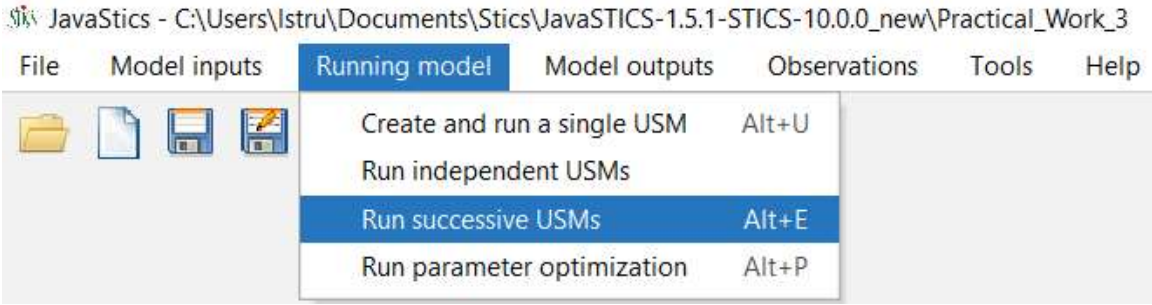
All the files necessary to run the simulations have been done for you! We are so kind, aren’t we?

First of all, you have to select the output data, in R or in JavaSTICS (use the “Model outputs > Choosing outputs > Reports” in javaSTICS):

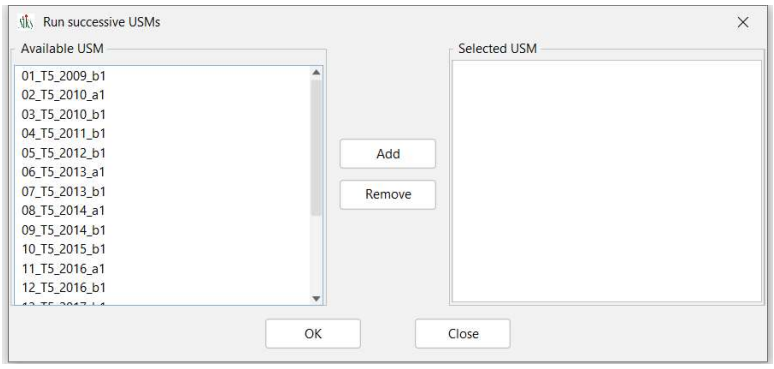
- masecnp = biomass of non-perennial organs in t DM ha^{-1}
- mafruit = biomass of fruits in t DM ha^{-1}
- mafauchetot = cumulative biomass of forage harvested during the USM in t DM ha^{-1}
- drat = cumulative amount of deep infiltration in mm
- Qles = cumulative amount of N leached at the base of the soil profile kg N ha^{-1}
- SOC = soil organic C in humus and zymogenous biomass until profhum depth in kg ha^{-1}
- SON = soil organic N in humus and zymogenous biomass until profhum depth in kg ha^{-1}
- SOCtot = total soil organic C in the residues and the entire soil profile in kg ha^{-1}
- SONtot = soil organic N in the residues and the entire soil profile in kg ha^{-1}

By phases at the end of each simulation (tick box “fin”).

To run the simulations, open JavaSTICS then select the workspace Practical_Work_3. Then do as follow:

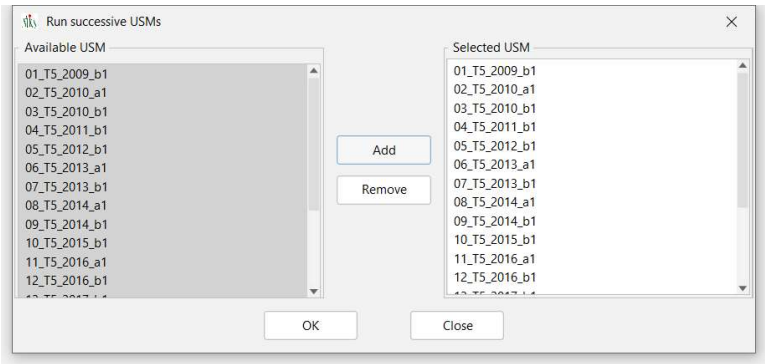


You should have this window:



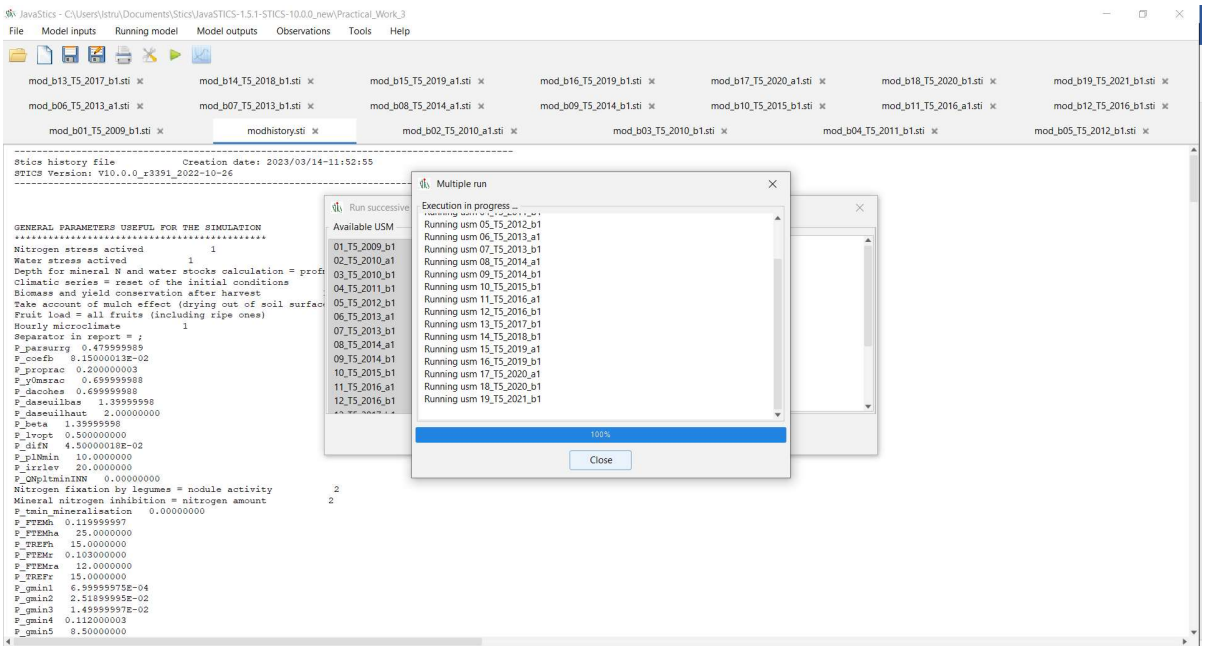
NB: USMs are sorted in the right order thanks to the numbers added before the name of the USM (01_...; 02_... etc). If you don’t do this, good luck to you!

Select all the available USMs then click on add. You should have this:



Now click on OK, then “Yes, I am sure!”

You should have this:



Let us have a look on output data:

You can see that the amount of water and mineral N in the soil are transmitted from an USM to another. The same thing is done for organic C&N pools which are transmitted from an USM to another.

If we look at alfalfa (USMs 12 and 13), we can also see that the plant N content is also transmitted from the first USM to the following one. This is the same thing for root system and plant biomass but you can see it only with daily outputs.

We can try to do the same using the R packages. Open the script “script_PW3.R”, adapt the path in the “Initializations” section of the script and run the first part of the script. Results are plotted in the file “dyn.png” in the workspace you defined.

II. Second part

Exercise:

We want to explore the effect of technical practices on the environment (organic C and N pool in the soil, cumulative nitrate lixiviation and water drainage at the end of the rotation) and on biomass production (crop biomass and yield). You can either use R or JavaSTICS.

<u>JavaSTICS:</u>

Please modify the USMs or the technical files by using the JavaSTICS interface.

1. Bare soil instead of catch crops, these farmers are called "we do what we want". Where do you have to do the modifications? What crop managements do you have to change?
   Some clues:
   - there is a bare soil (plant) file in the workspace, to simulate bare soil

2. Maximization of organic matter inputs to the soil: increasing period of soil cover by catch crop (until julian day 411 corresponding to mid-February) and no straw exportation for cash crops, these farmers are called "Organic matter lovers". Where do you have to do the modifications? What crop managements do you have to change?
   Some clues:
   - catch crop harvest dates are not calculated by the model (as these crops are not harvested at maturity) but are forced as phenological stages in the technical file
   - be careful not to plough your field before it has been harvested!
   - If you do not export straw, it means you let it as a "ressuite" after harvest 😉

3. Maximization of organic N inputs to the soil: only leguminous as CC and no exportation of the last cut of alfalfa, these farmers are called "Leguminous lovers". Where do you have to do the modifications? What crop managements do you have to change?
   Some clues:
   - vetch is a leguminous (as crimson clover), mustard is not.
   - cutting management of grasses is set in the "special techniques" part of the technical file

Please check the impact of these agricultural practices on:

the biomass production, the yield, the biomass produced by forages, the deep infiltration, N leaching and organic C and N pool at the end of each simulation since the simulation 01_T5_2009_b1. The work has already been done for the reference simulation, please do it for the new simulations with modified technical practices, then add the new outputs to the results.xlsx file.

What could you say about the impact of these agricultural practices?

R:

The work can be done by adapting the template given in the second part of the script "script_PW3.R" and creating new xls files (see R script comments) to handle the following cases:

1. Bare soil instead of catch crops, these farmers are called "we do what we want". Where do you have to do the modifications? What crop managements do you have to change?
   Some clues:
   - there is a bare soil (plant) file in the workspace, to simulate bare soil

2. Maximization of organic matter inputs to the soil: increasing period of soil cover by catch crop (until julian day 411 corresponding to mid-February) and no straw exportation for cash crops, these farmers are called "Organic matter lovers". Where do you have to do the modifications? What crop managements do you have to change?
   Some clues:
   - catch crop harvest dates are not calculated by the model (as these crops are not harvested at maturity) but are forced as phenological stages in the technical file
   - be careful not to plough your field before it has been harvested!
   - If you do not export straw, it means you let it as a "ressuite" after harvest 😉

3. Maximization of organic N inputs to the soil: only leguminous as CC and no exportation of the last cut of alfalfa, these farmers are called "Leguminous lovers". Where do you have to do the modifications? What crop managements do you have to change?
   Some clues:
   - vetch is a leguminous (as crimson clover), mustard is not.
   - cutting management of grasses is set in the "special techniques" part of the technical file

Please check the impact of these agricultural practices on:

the biomass production, the yield, the biomass produced by forages, the deep infiltration, N leaching and organic C and N pool at the end of each simulation since the simulation 01_T5_2009_b1.

After having generated the xls files and filled the corresponding sections of the R script, run it up to the end. Then, look at the files and plots generated by the "Post-treatments and plots" section of the script.

What could you say about the impact of the different agricultural practices studied?